| EXPT NO:6 | INTEGRATE DATA FROM VARIOUS SOURCES FOR COMPREHENSIVE ANALYSIS USING TABLEAU AND POWERBI |
|---|---|
| DATE: 24.07.2026 | |

**PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)**

1. **What is data integration in the context of Business Intelligence (BI)?**

2. **Explain the difference between a join and a data blend in Tableau.**

3. **What are relationships and cardinality (e.g., one-to-many, many-to-many) in Power BI's data model?**

4. **Why is data quality (e.g., consistent naming conventions, data types) crucial when integrating data from different sources?**

5. **Name two common challenges encountered during data integration.**

**IN-LAB EXERCISE:**

**OBJECTIVE:**
**To connect to, combine, and model data from multiple disparate sources in Tableau and Power BI to create a unified, comprehensive dashboard for holistic analysis.**
**PROCEDURE:**

1. **Data Collection:**
   - **Import data from at least two different sources (e.g., a CSV file and an Excel file).**
   - **Dataset 1: Sales Data (CSV file) with fields like Order ID, Order Date, Region, Sales Amount.**
   - **Dataset 2: Regional Sales Targets (Excel file) with fields like Region, Sales Target, Target Year.**
2. **Data Preparation and Transformation:**
   - **Clean both datasets: Check for inconsistencies in region names (e.g., "N. America" vs. "North America").**
   - **Ensure data types are correct (e.g., Sales Amount is numeric, Order Date is a date).**
   - **Handle any missing values as appropriate.**
3. **Data Integration and Modeling:**
   - **Tableau: Use the Data Source pane to establish a relationship or join between the Sales Data and Regional Targets tables using the common Region field.**
   - **Power BI: Use the Model view to create a relationship between the two tables, connecting them on the Region column. Define the cardinality (e.g., many-to-one).**
4. **Create Calculated Fields for Comparison:**
   - **Develop new measures that leverage both datasets.**

- **Examples: Target Achievement % = (SUM(Sales) / SUM(Sales Target)) * 100**
- **Sales vs. Target Variance = SUM(Sales) - SUM(Sales Target)**

5. **Design a Comprehensive Dashboard:**
   - **Create visuals that combine data from both sources to provide a single view of performance.**
   - **Add KPI cards for Total Sales and Overall Target Achievement %.**
   - **Use a gauge chart or bullet chart to show sales performance against targets for each region.**
   - **Create a combined bar chart showing Actual Sales vs. Target Sales by Region.**
   - **Use a map to visualize the Sales vs. Target Variance, highlighting underperforming or overperforming regions.**
6. **Validate Data Integrity:**
   - **Verify that the joined/related data is accurate by cross-checking values.**
   - **Ensure filters on the dashboard correctly update visuals that use data from both sources.**
7. **Publishing and Sharing:**
   - **Tableau: Publish the workbook to Tableau Public or Tableau Server.**
   - **Power BI: Publish the report to Power BI Service and share the link.**

**Experiment Datasets:**
**Dataset 1: Retail Sales Data (CSV)**
**Fields Used:**

- **Order ID – Unique identifier for each order**
- **Order Date – Date of the sales transaction**
- **Region – Sales region**
- **Sales – Sales revenue from the order**

**Dataset 2: Regional Sales Targets (Excel)**
**Fields Used:**

- **Region – Sales region (used as the key for joining/relationship)**
- **Sales Target – The sales goal for the region**
- **Target Year – The year for which the target is set**

**POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)**

1. **How did integrating the sales targets data provide a more complete analytical picture than using the sales data alone?**

2. **Describe the type of join or relationship you used to combine the two datasets and explain why it was the appropriate choice.**

3. **What challenges, if any, did you face regarding data granularity or mismatched keys (e.g., different region names) between the two sources?**

4. **How can creating variance calculations (e.g., Sales vs. Target) help a business manager make better decisions?**

5. **What other data source (e.g., marketing campaign data, customer demographics) could you integrate to further enrich this analysis, and what new insights might it provide?**

**RESULT:**

The retail sales and regional targets datasets were successfully connected, cleaned, and integrated in both Tableau and Power BI. By creating relationships between the sources, a unified data model was built. The resulting comprehensive dashboard visualized sales performance against set targets, enabling variance analysis and providing deeper, context-rich insights into regional performance that were not possible with a single data source.

**ASSESSMENT**

| Description | Max Marks | Marks Awarded |
|---|---|---|
| Pre Lab Exercise | 5 | |
| In Lab Exercise | 10 | |
| Post Lab Exercise | 5 | |
| Viva | 10 | |
| Total | 30 | |
| Faculty Signature | | |